



Math Level 5 explores topics such as arithmetic of fractions, decimals, and negative numbers. We also study geometry, number theory, statistics, and beginning algebra through lessons on 3D solids, factors and multiples, solving equations, statistics and data, sequences, and ratios. Problem-solving strategies are emphasized throughout the course, and each lesson includes challenging word problems and puzzles that guide students from the basics toward a deep understanding of the material.

Beast Academy provides a fun, rigorous, and challenging complete curriculum for aspiring math beasts. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through.

Book: Beast Academy 5A — 5D

Time Commitment: This 36-week course includes 105 minutes in-class and 60-90 minutes of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	3D Solids
2	Surface Area
3	Volume
4	Integer Arithmetic
5	Integer Exponents and Division
6	Algebraic Representation
7	Find x
8	Challenging Equations
9	Test on Book 5A
10	Median and Average
11	Mode, Range, & Data Display
12	Factors and GCF
13	LCM and Venn Diagrams
14	Factors and Relatively Prime
15	Fraction Addition & Subtraction
16	Fraction Multiplication
17	Fraction Division
18	Test on Book 5B

Week	Topic
19	Patterns and Sequences
20	Arithmetic Sequences, Triangular Numbers
21	Ratios
22	Rates
23	Unit Conversions
24	Decimal Multiplication
25	Decimal and Fraction Conversion
26	Decimal Division
27	Test on Book 5C
28	Introduction to Percent
29	Percents in Word Problems
30	Introduction to Square Roots
31	Tricky Square Roots
32	Square Roots and Geometry
33	Nonpositive Exponents & Rules
34	Power of a Power & More Rules
35	Exponents Review and Challenge
36	Test on Book 5D

Common Core State Standards:

The content covered in our Math Level 5 class is loosely based on the standard created by the Common Core State Standards Initiative. We cover many of the standards from grade 5, as well as standards for grades 4, 6, 7, and 8. You can find a document with the complete list on our Beast Academy Planning page (<https://beastacademy.com/resources/planning>). Or, you can access the standards by navigating directly to <https://beastacademy.com/u/pdfs/planning/standards/BA5Standards.pdf>

Sample Problems:

- How many **different** pairs of integers have a product of -128 ?

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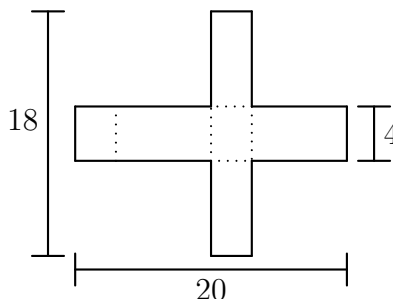
Suppose a represents the same **negative** number in each expression below. Order the expression from least to greatest.

$$a - 5a + 11a$$

$$-6a + 1 + 3a + 3$$

$$2a + 12a - 1$$

- What is the surface area of the rectangular prism with the net shown below?



- Natalie writes an arithmetic sequence in which the 20th term is 100 and the 50th term is 200. What is the 35th term of her sequence?
- When we write $\frac{5}{22}$ as a decimal, what is the 30th digit after the decimal point?
- Which two consecutive integers is $\frac{\sqrt{17}}{2}$ between?